Lesson notes · for you

Charts that tell the truth

Data science · Level 1 · Lesson 3

Time
30 min

Materials
A notebook and a pencil; A phone or tablet with Darsi, for the lab

Learning goals

- Choose bar, line, pie or scatter for a question.
- Spot a cut axis.
- Read a chart's title, axes and units.

Activity: Honest and tricky charts

1. Warm-up (5 min): Look at two charts of the same numbers and say which one exaggerates.
2. Draw (10 min): Draw a bar chart of your class's data with a scale from zero and all labels.
3. Trick (10 min): Draw the same data with a cut axis, then swap with a partner and find each other's trick.
4. Wrap-up (5 min): Say your finding in one sentence with a number in it, then fill in the exit ticket.

Discussion

- Why does the scale need to start at zero?
- Which chart fits your question?
- What labels did you nearly forget?

Check yourself

- Draw a tiny honest bar chart of two numbers, with a title and labelled axes.

Try it this week
Draw a bar chart of your family's favourite foods, with a title, labelled axes and a scale that starts at zero. Show it to someone and ask what they can tell from it.

Worksheet

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Name _____ Class _____ Date _____

1 **A poster's bar chart has an axis starting at 90. What should you check first?**

◯ The colours ◯ The title font ◯ Where the axis starts

2 **Match each question to the best chart.**

Which fruit is the favourite?	• • • •	Line chart
How did the temperature change this week?	• • • •	Pictogram
How many pets does each class have?	• • • •	Bar chart

3 **Your class voted: mango 4, apple 2. Put the steps for drawing a bar chart of the votes in order.**

☐ Draw the axes, with a scale that starts at 0

☐ Count the votes for each fruit

☐ Check that each bar matches its count

☐ Draw one bar for each fruit, as tall as its votes

4 **A bar chart's scale should start at zero.**

◯ True ◯ False

5 **What does every good chart need? Choose all that apply.**

☐ Labels on the axes ☐ A title ☐ Units ☐ Lots of colours

✎ **Have you seen a chart that made a small difference look huge? How did it do that?**

_____

_____

_____

Check your answers

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1 **Where the axis starts**

An axis that doesn't start at zero makes small differences look big.

2 **Which fruit is the favourite? — Bar chart**
How did the temperature change this week? — Line chart
How many pets does each class have? — Pictogram

Bars compare, lines show change, pictures make counts friendly.

3 **Top to bottom: 2, 1, 4, 3**

You need the counts first, to choose the scale.

4 **True**

If it starts higher, small differences look much bigger than they are.

5 **A title; Labels on the axes; Units**

Title, labels and units tell people what they are looking at.